波动方程　平面波.

真空中

$$\vec{\rho} = 0, \quad \vec{J} = 0$$

其中, Maxwel eqs

$$\nabla \cdot \vec{D} = \rho = 0$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$\nabla \cdot \vec{B} = 0$$

考虑,

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times (\nabla \times \vec{E}) = \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\nabla^2 \vec{E}$$

$$= -\nabla \times (\frac{\partial \vec{B}}{\partial t})$$

$$= -\frac{\partial}{\partial t}(\nabla \times \vec{B})$$

$$= -\frac{\partial}{\partial t}(\mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t})$$

$$= -\mu_0 \varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

即

$$\nabla^2 \vec{E} - \mu_0 \varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} = 0$$

令 $c \equiv \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$ 为真空中光速，有：

波动方程：

$$\nabla^2 \vec{E} - \frac{1}{c^2} \frac{\partial^2 \vec{E}}{\partial t^2} = 0$$

类似地，

$$\nabla^2 \vec{B} - \frac{1}{c^2} \frac{\partial^2 \vec{B}}{\partial t^2} = 0$$

绝缘介质中情形

$$\vec{D} = \varepsilon \vec{E}$$

$$\vec{H} = \frac{\vec{B}}{\mu}$$

强调，

$$\varepsilon = \varepsilon(\omega) \quad , \quad \mu = \mu(\omega)$$

为频率的函数.

对于单色波 —— 频率单一的波.

(定态波 —— 频率单一的简谐波)

此时

$$\rho = 0 \qquad \vec{J} = 0$$

$$\nabla \times (\nabla \times \vec{H}) = \nabla \times (\frac{\partial \vec{D}}{\partial t})$$

$$\nabla (\nabla \cdot \vec{H}) - \nabla^2 \vec{H} = \frac{\partial}{\partial t} (\nabla \times \vec{D})$$

$$\frac{1}{\mu_0} \nabla (\nabla \cdot \vec{H}) - \frac{1}{\mu_0} \nabla^2 \vec{B} = \varepsilon \frac{\partial}{\partial t} (-\frac{\partial \vec{B}}{\partial t})$$

$$\nabla^2 \vec{B} - \frac{1}{v^2} \frac{\partial^2 \vec{B}}{\partial t^2} = 0$$

其中，

$$v \equiv \frac{1}{\sqrt{\varepsilon \mu}}$$

为绝缘介质中光速

进一步地

$$\varepsilon = \varepsilon_0 \varepsilon_r$$

$$\mu = \mu_0 \mu_r$$

$$v = \frac{1}{\sqrt{\mu_0 \varepsilon_0 \mu_r \varepsilon_r}} = \frac{c}{n}$$

其中，

$$n \equiv \sqrt{\varepsilon_r \mu_r}$$

称材料的**折射率**.

综上，

$$\nabla^2 \vec{B} - \frac{1}{v^2} \frac{\partial^2 \vec{B}}{\partial t^2} = 0$$

$$\nabla^2 \vec{E} - \frac{1}{v^2} \frac{\partial^2 \vec{E}}{\partial t^2} = 0$$

在定态波情形下，

$$\vec{E}(\vec{x}, t) = \vec{E}(\vec{x}) \cos \omega t$$

$$= \vec{E}(\vec{x}) e^{-i\omega t}$$

约定，只有实部有物理意义.

同样地，

$$\vec{B}(\vec{x}, t) = \vec{B}(\vec{x}) e^{-i\omega t}$$

$$\nabla^2 \vec{E}(\vec{x},t) = \nabla^2 \vec{E}(\vec{x})\, e^{-i\omega t}$$

$$\frac{\partial^2 \vec{E}(\vec{x},t)}{\partial t^2} = \frac{\partial^2}{\partial t^2}\left(\vec{E}(\vec{x})\, e^{-i\omega t}\right)$$

$$= \vec{E}(\vec{x})\,(-i\omega)^2\, e^{-i\omega t} = -\omega^2 \vec{E}(\vec{x},t)$$

即:

$$e^{i\omega t}\ \nabla^2 \vec{E}(\vec{x}) = -\omega^2\, e^{i\omega t}\ \vec{E}(\vec{x})$$

$$\nabla^2 \vec{E}(\vec{x}) + \omega^2 E(\vec{x}) = 0$$

dim

$$\vec{k} \equiv \frac{\omega}{v}\vec{n} = \frac{2\pi}{\lambda}\vec{n}$$

为 波矢, 其中 $\vec{n}$ 为波传播方向.

于是代入有:

Helmholtz eq

$$\nabla^2 \vec{E}(\vec{x}) + k^2 \vec{E}(\vec{x}) = 0$$

辅助: $k^2 = \omega^2 \mu\varepsilon$ 条件: $\nabla \cdot \vec{E}(\vec{x}) = 0$

$$\nabla^2 \vec{B}(\vec{x}) + k^2 \vec{B}(\vec{x}) = 0$$

辅助: $k^2 = \omega^2 \mu\varepsilon$ 条件: $\nabla \cdot \vec{B}(\vec{x}) = 0$

另外，由 Faraday's law：

$$\nabla \times \vec{E}(\vec{x}, t) = -\frac{\partial \vec{B}(\vec{x}, t)}{\partial t}$$

$$\nabla \times (e^{-i\omega t}\vec{E}(\vec{x})) = -\frac{\partial}{\partial t}(e^{-i\omega t}B(\vec{x}))$$

$$= i\omega \vec{B}(\vec{x}) e^{-i\omega t}$$

$$= e^{-i\omega t} \nabla \times \vec{E}(\vec{x})$$

$$\vec{B}(\vec{x}) = -i\frac{1}{\omega} \nabla \times \vec{E}(\vec{x})$$

同理，由安培定律，

自由空间中平面电磁波

Helmholtz eq 最简单的解 —— 平面波解.

$$\vec{E}(\vec{x},t) = \vec{E}(\vec{x})\,e^{-i\omega t} = \vec{E}_0\,e^{i(\vec{k}\cdot\vec{x}-\omega t)}$$

$$\vec{E}(\vec{x}) = \vec{E}_0\,e^{i\vec{k}\cdot\vec{x}}$$

dim. $\phi = \vec{k}\cdot\vec{x} - \omega t$ 称相位

描述波在 t 时刻 x 处状态.

验证:

$$\nabla^2\vec{E} = \nabla^2\left(\vec{E}_0 e^{i(\vec{k}\cdot\vec{x}-\omega t)}\right) = \vec{E}_0\,e^{i\omega t}\left(\nabla^2 e^{i\vec{k}\cdot\vec{x}}\right)$$

$$\nabla^2 e^{i\vec{k}\cdot\vec{x}} = \nabla\cdot\left(\nabla e^{i\vec{k}\cdot\vec{x}}\right)$$

而

$$\nabla e^{i\vec{k}\cdot\vec{x}} = \frac{\partial}{\partial x}\left(e^{i\vec{k}\cdot\vec{x}}\right)\hat{e}_x + \frac{\partial}{\partial y} + \cdots$$

不严格推导!

$$\neq \frac{\partial}{\partial x} e^{i\vec{k}\cdot\vec{x}} = i\vec{k}\,e^{i\vec{k}\cdot\vec{x}}$$

$$\nabla\cdot\left(\nabla e^{i\vec{k}\cdot\vec{x}}\right) = \nabla\cdot\left(i\vec{k}\,e^{i\vec{k}\cdot\vec{x}}\right) = \underbrace{i\left(\nabla\cdot\vec{k}\right)}_{=0} e^{i\vec{k}\cdot\vec{x}} + i\vec{k}\cdot\nabla e^{i\vec{k}\cdot\vec{x}}$$

$$= (ik)^2 e^{i\vec{k}\cdot\vec{x}} = -k^2 e^{i\vec{k}\cdot\vec{x}}$$

得证 ∥

综上

$$\vec{E} = \vec{E}_0 e^{i(\vec{k}\cdot\vec{x} - \omega t)}$$

且要满足，

$$\nabla \cdot \vec{E}_0 e^{i(\vec{k}\cdot\vec{x} - \omega t)} = 0$$

$$\vec{E}_0 e^{-i\omega t} \nabla \cdot e^{i\vec{k}\cdot\vec{x}} = 0$$

$$\vec{E}_0 e^{-i\omega t} (i\vec{k} \cdot e^{i\vec{k}\cdot\vec{x}}) = 0$$

即

$$i\vec{k} \cdot \vec{E}(\vec{x}, t) = 0$$

$$\vec{k} \perp \vec{E}(\vec{x}, t)$$

- 平面电磁波一定为横波，没有纵波解.
- 同理，$\vec{k} \cdot \vec{B}(\vec{x}, t) = 0$，$\vec{k} \perp \vec{B}$

进一步，

$$\vec{B}(\vec{x}) = -\frac{i}{\omega} \nabla \times \vec{E}_0 e^{i\vec{k}\cdot\vec{x}}$$

$$= -\frac{i}{\omega} \nabla e^{i\vec{k}\cdot\vec{x}} \times \vec{E}_0$$

$$= -\frac{i}{\omega} (i\vec{k} e^{i\vec{k}\cdot\vec{x}}) \times \vec{E}_0$$

$$= \frac{\vec{k}}{\omega} \times \vec{E}(\vec{x})$$

总结：

(1) 平面波是个横波.

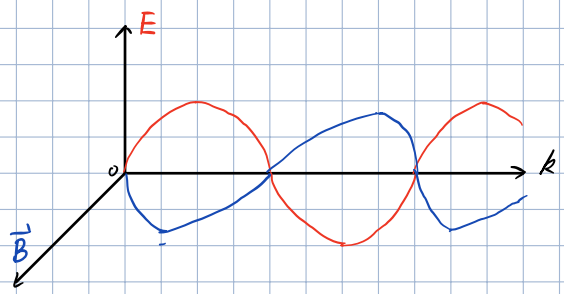
$$\vec{E}\cdot\vec{k}=0 \qquad \vec{B}\cdot\vec{k}=0$$

(2) $\vec{B}\perp\vec{E}$

$$\vec{B}(\vec{x})=\frac{\vec{k}}{\omega}\times\vec{E}(\vec{x})$$

$\vec{E}$、$\vec{B}$、$\vec{k}$ 三者互相垂直.

(3) $\vec{E}$、$\vec{B}$ 同相位.



(4) 对均匀介质.

$$\frac{|E|}{|B|}=\frac{|E|}{\left|\frac{\vec{k}}{\omega}\right||\vec{E}|}$$

$$=\frac{\omega}{k}=\frac{1}{\sqrt{\varepsilon\mu}}=\frac{c}{n}$$

(5) $k=\omega\sqrt{\mu\varepsilon}$

$$v=\frac{\omega}{k}=\frac{c}{n}$$

(6) 偏振 —— 约定为电场 E 的振动方向

电磁波的能量.

1. 瞬时值.

$$\omega = \frac{1}{2}\vec{E}\cdot\vec{D} + \frac{1}{2}\vec{B}\cdot\vec{H} = \frac{1}{2}\varepsilon E^2 + \frac{1}{2}\frac{1}{\mu}B^2$$

$$B^2 = \mu\varepsilon E^2 = \qquad \varepsilon E^2 = \frac{1}{\mu}B^2$$

电磁波电场与磁场储存能量相同.

$$\omega = \varepsilon E_0^2 \cos^2(\vec{k}\cdot\vec{x} - \omega t)$$

dim $\quad \vec{S} = \frac{1}{\mu}\vec{E}\times\vec{B} = \frac{1}{\mu}\vec{E}\times(\frac{\vec{k}}{\omega}\times\vec{E})$

$$= \frac{1}{\mu\omega}[\,E^2\vec{k} - (\vec{E}\cdot\vec{k})\vec{E}\,] = \frac{\vec{k}}{\mu\omega}E^2$$

$$= \sqrt{\frac{\varepsilon}{\mu}}E_0^2\cos^2(\vec{k}\cdot\vec{x} - \omega t)\,\vec{n}$$

瞬时值是无法测量的

2. 平均值.

$$\overline{\omega} = \frac{\int_0^T \omega\,dt}{T} = \frac{1}{T}\varepsilon E_0^2\int_0^T \cos^2(\vec{k}\cdot\vec{x} - \omega t)\,dt$$

$$= \frac{1}{2}\varepsilon E_0^2 = \frac{1}{2}\omega_{max}.$$

$$\overline{S} = \frac{1}{T}\int_0^T \vec{S}\,dt = \frac{S_{max}}{2}$$

导体中电磁波.

$\vec{j} = \sigma \vec{E}$

$E \longrightarrow \vec{J} \longrightarrow$ 热量　　E 越来越弱

耗散.

问题：本构关系.

静态时，$\vec{j} = \sigma \vec{E}$　各时时？

电场，

$$\vec{E}(\vec{r}, t) = \vec{E}(\vec{r}) e^{-i\omega t}$$

电流也应各时变化，频率相同.

$$\vec{j}(\vec{r}, t) = \vec{j}(\vec{r}) e^{-i\omega t}$$

考虑某个电子受电场与阻尼作用.

$$\vec{F} = e\vec{E} - k\vec{V} = e\vec{E}_0 e^{-i\omega t} - \frac{m}{\tau}\vec{V} = m\frac{d\vec{V}}{dt}$$

电场　阻尼.

$$\vec{v}_0 = \frac{-e\vec{E}_0}{im(\omega + \frac{i}{\tau})}$$

$$\vec{j} = e n_e \vec{v} = \underbrace{\frac{-ne^2}{im(\omega+\frac{i}{\tau})}}_{\sigma} e^{-i\omega t}\vec{E}_0$$

电导率为1个复数.

$\sigma = (\underbrace{Re\,\sigma}, \underbrace{Im\,\sigma})$

耗散.　电场能量 $\longrightarrow$ 电荷动能

$$= \frac{-ne^2\vec{E}_0}{im(\omega+\frac{i}{\tau})}$$

Im　Re

对好的导体　$\tau \sim 10^{-14}s$

① $\omega \gg \frac{1}{\tau}$　无耗散，光可进入.

② $\omega \ll \frac{1}{\tau}$　耗散

↓

近似可见光.

$$\frac{1}{\mu_0}\nabla\times\vec{B} = \vec{j} + \frac{\partial\vec{E}}{\partial t}\varepsilon_0 = \sigma\vec{E} + \frac{\partial\vec{E}}{\partial t}\varepsilon_0$$

$$\vec{E} = \vec{E}(\vec{r})e^{-i\omega t}$$

$$\frac{1}{\mu_0}\nabla\times\vec{B} = \sigma\vec{E}(\vec{r})e^{-i\omega t} - i\varepsilon_0\omega\vec{E}(\vec{r})e^{-i\omega t} = [\sigma - i\omega_0\varepsilon(\omega)]\vec{E}$$

$$\nabla\times\vec{H} = -i\omega\varepsilon(\omega)\vec{E}$$

其中，

$$\varepsilon(\omega) = [1 + i\frac{\sigma(\omega)}{\varepsilon_0\omega}]\varepsilon_0$$

本构关系使用.

$$\vec{j}(\omega) = \sigma(\omega)\vec{E}$$

$$\nabla\cdot\vec{E} = \frac{\rho}{\varepsilon_0}$$

$$\nabla\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} \quad ②$$

$$\nabla\cdot\vec{B} = 0 \quad ③$$

$$\nabla\times\vec{B} = \mu_0(\vec{j} + \varepsilon\frac{\partial\vec{E}}{\partial t})$$

考虑连续性方程：

$$\frac{\partial\rho}{\partial t} + \nabla\cdot\vec{j} = 0$$

$$\frac{\partial \rho}{\partial t} = \varepsilon_0 \nabla \cdot \frac{\partial \vec{E}}{\partial t} = -\nabla \cdot \vec{j}$$

$$\nabla \cdot \vec{E} = \frac{1}{\varepsilon_0} \sigma(\omega) \nabla \cdot \vec{E}$$

$$\nabla \cdot (\varepsilon(\omega) \vec{E}) = 0 \qquad (1)$$

$$\nabla \times (\vec{H}) = \nabla \times (\mu_0 \vec{B}) = \varepsilon(\omega) \vec{E} \qquad (4)$$

和上面相同方法：

$$\nabla \times \nabla \times \vec{H} = \nabla \times (-i\omega \varepsilon(\omega) \vec{E})$$

$$\vec{H} = H_0 e^{-i\vec{k}\cdot\vec{r}} e^{-i\omega t}$$

$$\varepsilon_r(\omega) = 1 + i r \frac{\sigma(\omega)}{\varepsilon_0 \omega}$$

$$= 1 - \frac{\omega_p^2}{\omega(\omega + \frac{i}{\tau})}$$

$$\omega_p = \sqrt{\frac{n e^2}{\varepsilon_0 m}}$$

电磁波在介质界面上的反射与折射.

- 入射角、反射角、折射角关系.
- 入射波、反射波、折射波的振幅比与相对相位.

基础： E、B 的边值关系

$$e_n \times (E_2 - E_1) = 0$$

$$e_n \times (H_2 - H_1) = \alpha$$

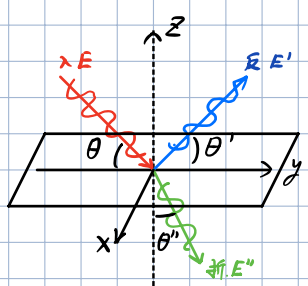
只考虑这两式.

$$e_n \cdot (D_2 - D_1) = \sigma$$

$$e_n \cdot (B_2 - B_1) = 0$$

假设： 入射、反射、折射波均为平面波.

1. 反射定律与折射定律



入射 $E = E_0 e^{i(\vec{k}\cdot\vec{x} - \omega t)}$

反射 $E' = E_0' e^{i(\vec{k}'\cdot\vec{x} - \omega t)}$

折射 $E'' = E_0'' e^{i(\vec{k}''\cdot\vec{x} - \omega t)}$

$$e_n \times (E_0 e^{i\vec{k}\cdot\vec{x}} - E_0' e^{i\vec{k}'\cdot\vec{x}}) = E_0'' e^{i\vec{k}''\cdot\vec{x}}$$

选平面 $z=0$，上式应对 $\forall x, y$ 成立.

$$\vec{k}\cdot\vec{x} = \vec{k}'\cdot\vec{x} = \vec{k}''\cdot\vec{x}$$

$$\Rightarrow \quad k_x = k_x' = k_x'' \ , \quad k_y = k_y' = k_y''$$

取入射波在xz平面上

$$k_y = 0 = k_y' = k_y''$$

以 θ、θ'、θ'' 分别表入射角、反射角、折射角.

$$k_x = k\sin\theta, \quad k_x' = k'\sin\theta', \quad k_x'' = k''\sin\theta''$$

以 v_1、v_2 表电磁波在2介质中的相速.

$$k = k' = \frac{\omega}{v_1} \qquad k'' = \frac{\omega}{v_2}.$$

代入，有:

$$\theta = \theta'$$

$$\frac{\sin\theta}{\sin\theta''} = \frac{v_1}{v_2}$$

对电磁波:

$$v = \frac{1}{\sqrt{\mu\varepsilon}}$$

$$\frac{\sin\theta}{\sin\theta''} = \frac{\sqrt{\mu_2\varepsilon_2}}{\sqrt{\mu_1\varepsilon_1}} = n_{21}.$$

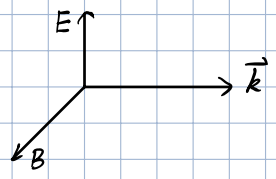
n_{21}: 折射率.

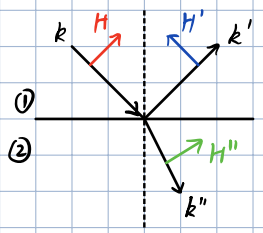
对隔铁磁反射一般介质，

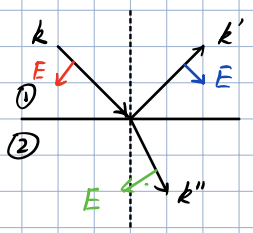
$$\mu \approx \mu_0$$

$$n_{21} = \sqrt{\frac{\varepsilon_2}{\varepsilon_1}}.$$

2. 振幅关系　菲涅尔公式







$$\begin{cases} E \perp \text{入射面} & (1) \\ E \parallel \text{入射面}. & (2) \end{cases}$$

(1) $E \perp$ 入射面.

当界面上自由电流密度 $\alpha = 0$

$$E + E' = E''$$

$$H\cos\theta - H'\cos\theta' = H''\cos\theta''$$

又

$$H = \sqrt{\frac{\varepsilon}{\mu}}E$$

对非铁磁性介质:

$$\mu = \mu_0$$

$$\therefore \quad \sqrt{\varepsilon_1}(E - E')\cos\theta = \sqrt{\varepsilon_2}E''\cos\theta''$$

利用折射定律, 有:

$$\frac{E'}{E} = \frac{\sqrt{\varepsilon_1}\cos\theta - \sqrt{\varepsilon_2}\cos\theta''}{\sqrt{\varepsilon_1}\cos\theta + \sqrt{\varepsilon_2}\cos\theta''} = -\frac{\sin(\theta-\theta'')}{\sin(\theta+\theta'')}$$

$$\frac{E''}{E} = \frac{2\sqrt{\varepsilon_1}\cos\theta}{\sqrt{\varepsilon_1}\cos\theta + \sqrt{\varepsilon_2}\cos\theta''} = \frac{2\cos\theta\sin\theta''}{\sin(\theta+\theta'')}$$

$$\nabla\times\vec{E} = \mu\frac{\partial\vec{H}}{\partial t} = \mu(i\omega)\vec{H}$$

$$= ik\vec{E}$$

$$H = \frac{k}{\omega}\cdot\frac{1}{\mu}E = \frac{\omega}{v\omega}\frac{1}{\mu}E$$

$$= \frac{1}{v}\frac{1}{\mu}E = \frac{\sqrt{\mu\varepsilon}}{\mu}E$$

$$= \sqrt{\frac{\varepsilon}{\mu}}E.$$

(2) E // 入射面

边值关系：

$$E\cos\theta - E'\cos\theta = E''\cos\theta''$$

$$H + H' = H''$$

仍然，将 H 用 E 表示：

$$\sqrt{\varepsilon_1}(E + E') = \sqrt{\varepsilon_2}E''$$

利用折射定律：

$$\frac{E'}{E} = \frac{\tan(\theta-\theta'')}{\tan(\theta+\theta'')}$$

$$\frac{E''}{E} = \frac{2\cos\theta\sin\theta''}{\sin(\theta+\theta'')\cos(\theta-\theta'')}$$

综上，我们有：

Fresnel 公式：

$E\perp$ 入射面：

$$\frac{E'}{E} = -\frac{\sin(\theta-\theta'')}{\sin(\theta+\theta'')}$$

$$\frac{E''}{E} = \frac{2\cos\theta\sin\theta''}{\sin(\theta+\theta'')}$$

E // 入射面：

$$\frac{E'}{E} = \frac{\tan(\theta-\theta'')}{\tan(\theta+\theta'')}$$

$$\frac{E''}{E} = \frac{2\cos(\theta-\theta'')}{\sin(\theta+\theta'')\cos(\theta-\theta'')}$$

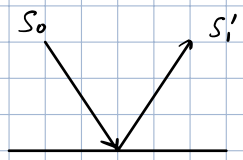
- ⊥入射面偏振的波与//入射面偏振的波的折射与反射行为不同.
- 若入射波为自然光(2种偏振光等量混合),经折射、反射后,由于2偏振分量的反射波与折射波强度不同,反射波与折射波都变为部份偏振光.
- 在 $\theta+\theta''=90^\circ$ 特殊情形下,E//入射面分量无反射波,反射光变为⊥入射面偏振的完全偏振光.

  Brewster's law.
- 入射波、反射波,折射波的相位关系:

  E⊥入射面, 当 $\varepsilon_2>\varepsilon_1$ 时, $\theta>\theta''$ $\frac{E'}{E}<0$

  反射波电场与入射波电场反相,此现象称反射过程中的半波损失.

材料透射率.



以 P 波为例:

定义 反射率 r

$$r = \frac{\langle \vec{S}_1 \rangle \cdot \hat{e}}{\langle \vec{S}_0 \rangle \cdot \hat{e}} = \left| \frac{Z_2 \cos\theta - Z_1 \cos\theta}{Z_2 \cos\theta + Z_1 \cos\theta} \right|$$

定义 透射率 T

$$T = \frac{\langle \vec{S}_2 \rangle \cdot \hat{e}}{\langle \vec{S}_0 \rangle \cdot \hat{e}} = \frac{|E_2|^2 \cos\theta' / Z_2}{|E_0|^2 \cos\theta / Z_1}$$

$$r_s = \left| \frac{Z_2 \cos\theta - Z_1 \cos\theta''}{Z_2 \cos\theta + Z_1 \cos\theta''} \right|^2$$

$$r_p = \left| \frac{Z_2 \cos\theta'' - Z_1 \cos\theta}{Z_2 \cos\theta'' + Z_1 \cos\theta} \right|^2$$

Brewster's law

考虑非磁性介质.

$$\mu \longrightarrow \mu_0$$

$$\frac{\sin\theta}{\sin\theta''} = \frac{n_2}{n_1} = \frac{\sqrt{\varepsilon_2}}{\sqrt{\varepsilon_1}} = \frac{z_1}{z_2}$$

$$r_s = \left| \frac{z_2\cos\theta - z_1\cos\theta''}{z_2\cos\theta + z_1\cos\theta''} \right| = \frac{\sin^2(\theta-\theta'')}{\sin^2(\theta+\theta'')}$$

$$r_p = \left| \frac{z_2\cos\theta'' - z_1\cos\theta}{z_2\cos\theta'' + z_1\cos\theta} \right| = \frac{\tan^2(\theta-\theta'')}{\tan^2(\theta+\theta'')}$$

$$\theta + \theta'' = \frac{\pi}{2} \quad \Longrightarrow \quad r_p = 0$$

$\theta + \theta'' = \frac{\pi}{2}$ 时的角称为 Brewster 角.

全反射

Snell's law

$$\frac{\sin\theta}{\sin\theta''} = \frac{n_1}{n_2}$$

$$\sin\theta'' = \frac{n_1}{n_2}\sin\theta$$

if $n_1 > n_2$ $\quad \frac{n_1}{n_2} > 1$

when $\sin\theta > \frac{n_2}{n_1}$ 即 $\theta > \arcsin\frac{n_2}{n_1}$

$$\theta'' \longrightarrow 0$$

这是几何光学下的结论.

问题：全反射中，电磁场情形如何？

$$E = \vec{E_0} e^{i(\vec{k}\cdot\vec{x} - \omega t)}$$

$$E' = E_0' e^{i(\vec{k}'\cdot\vec{x} - \omega t)}$$

$$E'' = E_0'' e^{i(\vec{k}''\cdot\vec{x} - \omega t)}$$

边值关系：

$$k_x' = k_x = k\sin\theta$$

$$k'' = k\frac{v_1}{v_2} = kn_{21}$$

when $\sin\theta > n_{21}$.

$$k_x'' > k''$$

$$k_z'' = \sqrt{k''^2 - k''^2_x} = ik\sqrt{\sin^2\theta - n_{21}^2}$$

是一个虚数.

折射波情况如何?

令

$$k_z'' = iK \qquad K = k\sqrt{\sin^2\theta - n_{21}^2}$$

折射波电场表示式:

$$E'' = E_0'' e^{-Kz} e^{i(k_x'' x - \omega t)}$$

when $z \to -\infty$, $E'' \to \infty$. 上式所表示的波不能存在于全空间, 只存在于 $z>0$ 的半空间

E'' 场强大小随 z 指数衰减. 这种电磁波只存在于界面附近一薄层内. 称表面波.

该层厚度 $\sim K^{-1}$

$$K^{-1} = \frac{1}{k\sqrt{\sin^2\theta - n_{21}^2}} = \frac{\lambda_1}{2\pi\sqrt{\sin^2\theta - n_{21}^2}}$$

考虑磁场强度, $E''\perp$ 入射面情形 ($E'' = E_y''$).

$$H_z'' = \sqrt{\frac{\varepsilon_2}{\mu_2}}\frac{k_x''}{k''}E_y'' = \sqrt{\frac{\varepsilon_2}{\mu_2}}\frac{\sin\theta}{n_{21}}E''$$

$$H_x'' = -\sqrt{\frac{\varepsilon_2}{\mu_2}}\frac{k_z''}{k''}E_y'' = -i\sqrt{\frac{\varepsilon_2}{\mu_2}}\sqrt{\frac{\sin^2\theta}{n_{21}^2} - 1}\ E''$$

能流：

$$S_x'' = \frac{1}{2} Re\left(E_y'' * H_z''\right) = \frac{1}{2}\sqrt{\frac{\varepsilon_2}{\mu_2}}\,|E_0''|^2 e^{-2Kz}\frac{\sin\theta}{n_{21}}$$

$$S_z'' =$$

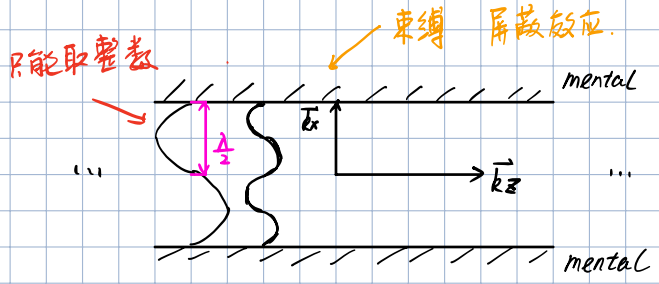
波导与谐振腔

有界空间中的电磁波

无界空间中，电磁波最基本存在形式：平面波.

这种波电场与磁场都做横向振荡，称横电磁(TEM)波

波导：



受约束的平面波

边界条件：

$\vec{k}_x$ 被约束 ， $\vec{k}_z$ 不受约束。

① 边界条件：

$$\vec{n}\cdot\vec{B}=0$$

$$\vec{n}\times\vec{B}=0$$

② eg：

$$\left(\nabla^2-\frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right)\begin{pmatrix}\vec{E}\\ \vec{B}\end{pmatrix}=0$$

单色光：

$$\vec{E}(\vec{r},t)=\vec{E}_0(x,y)\,e^{-i\omega t}\,e^{ik_z\cdot z}$$

代入：

$$\left(\partial_x^2+\partial_y^2+\partial_z^2-\frac{1}{c^2}\partial_t^2\right)\vec{E}(r,t)=0$$

$$\left(\partial_x^2+\partial_y^2-k_z^2+\frac{\omega^2}{c^2}\right)\vec{E}_0(x,y)=0.$$

令 $k_c^2 = \frac{\omega^2}{c^2} - k_z^2$，化为：

$$(\partial_x^2 + \partial_y^2 + k_c^2)\,\vec{E}_0(x,y) = 0$$